

2. (3.25 points)

Given the following information (\$000,000):

Accident <u>Year</u>	Cumulative Reported Loss <u>@ 24 Months</u>	Ultimate <u>Loss</u>
2011	36	75
2012	40	71
2013	35	64
2014	25	

a. (1.25 points)

Using the least squares method, estimate ultimate loss for accident year 2014.

b. (0.5 point)

For each of the following scenarios, briefly describe a potential problem with the output of the least-squares method:

- i. The slope parameter is negative.
- ii. The intercept parameter is negative.

c. (1.5 points)

Due to a regulatory change, the following is anticipated:

- No change in the reporting pattern.
- Standard deviation of reported loss as of 24 months will be 10% of estimated ultimate loss.
- Expected ultimate loss for accident year 2014 will decrease 20%.
- Standard deviation of accident year 2014 ultimate loss is expected to be \$6,000,000.

Using the Bayesian credibility method, estimate the revised ultimate loss for accident year 2014.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 2	
TOTAL POINT VALUE: 3.25	LEARNING OBJECTIVE: A1: Calculate unpaid claim estimates using credibility models.
SAMPLE ANSWERS	
Part a: 1.25 points	
<i>Sample Answer 1</i> $\bar{X} = \frac{36 + 40 + 35}{3} = 37$ $\bar{Y} = \frac{75 + 71 + 64}{3} = 70$ $\overline{XY} = \frac{36 * 75 + 40 \times 71 + 35 \times 64}{3} = 2593.33$ $\overline{X^2} = \frac{36^2 + 40^2 + 35^2}{3} = 1373.67$ $b = \frac{\overline{XY} - \bar{X}\bar{Y}}{\overline{X^2} - \bar{X}^2} = 0.713$ $a = \bar{Y} - b \times \bar{X} = 43.62$ $2014 \text{ Ultimate Loss} = a + b \times 25 = 61.45$ <i>Sample Answer 2</i> $r = 0.33$ $\sigma_X = 2.16, \sigma_Y = 4.55$ $a = \frac{r \times \sigma_Y}{\sigma_X} = 0.7$ $b = \bar{Y} - a \times \bar{X} = 43$ $U_{2014} = 25 \times 0.7 + 43 = 61$	
Part b: 0.5 point	
<i>Sample Answer 1</i> i) If $b < 0$, then y decreases as x increases. ii) If $a < 0$, then y is negative for small values of x. <i>Sample Answer 2</i> i) If b is negative, then ultimate loss (Y) decreases when reported loss (x) increases. ii) If a is negative, then ultimate loss (Y) is negative when reported loss (x) is zero.	

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 3

- i) Ult loss might be negative when intercept is small/reported is large.
- ii) Ult loss might be negative when reported loss is small/slope is small.

Sample Answer 4

- i) Negative slope suggests negative development.
- ii) Negative intercept suggests when there have been no reported losses, the ultimate is negative.

Part c: 1.5 point

Sample Answer 1

$$L(x) = Z \times \frac{X}{d} + (1 - Z)E(y)$$

$$E(y) = (1 - .2) \times \frac{75 + 71 + 64}{3} = 56$$

$$X = 25$$

$$d = \frac{36 + 40 + 35}{75 + 71 + 64} = .52857$$

$$Z = \frac{VHM}{VHM + EVPV} = \frac{10.05}{10.05 + 31.72} = .2407$$

$$VHM = E^2\left(\frac{X}{Y}\right) Var(Y) = (.52857)^2(6)^2 = 10.05796$$

$$EVPV = Var\left(\frac{X}{Y}\right) [Var(Y) + E^2(Y)] = .1^2(6^2 + 56^2) = 31.72$$

$$L(x) = (.2407) \times \frac{25}{.52857} + (1 - .2407)(56) = 53.905 \text{ M}$$

Sample Answer 2

$$\sigma_d = 0.1$$

$$Y = 0.8 \times 70 = 56$$

$$\sigma_Y = 6$$

$$d = \frac{37}{70} = 0.5286$$

$$VHM = \sigma_Y^2 d^2 = 6^2(0.5286)^2 = 10.058$$

$$EVPV = \sigma_d^2[\sigma_Y^2 + Y^2] = 0.1^2(6^2 + 56^2) = 31.72$$

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

$$Z = \frac{VHM}{VHM + EVPV} = \frac{10.058}{10.058 + 31.72} = .2407$$

$$L = .2407 \left(\frac{25}{.5286} \right) + (1 - .2407)(56) = 53.904$$

EXAMINER'S REPORT

Part a

- In general, candidates performed very well on this subpart.
- A variety of different answers were accepted for *a*, *b*, and 2014 ultimate loss, due to rounding differences in the calculations of X-Bar, Y-Bar, etc.
- The most common mistakes included minor calculation errors.
- Numerous candidates performed the least squares regression using their calculators and used the output to calculate the estimated ultimate loss for 2014. This was acceptable, if performed correctly. However, if no work was shown and the ultimate value was calculated incorrectly, only minimal partial credit was awarded.
- It was not necessary to indicate that the dollar amounts were in millions.

Part b

Candidates were expected to identify potential issues with the least squares development method when either the estimated slope or intercept parameters are negative. In general, most candidates performed very well on this question. Candidates that received full credit made a clear connection between the negative parameter estimate and possibly inappropriate results. Conversely, simply stating that a "negative intercept parameter could lead to negative ultimate loss" did not receive credit because there was no explanation of the scenarios when the ultimate loss would be negative.

Candidates were not required to state a solution to either potential problem. No credit was awarded or deducted for including possible solutions to the stated problems. If candidates only included potential solutions but did not explicitly address why the negative parameters might be inappropriate, they received no credit.

Candidates also lost credit when they did not clearly differentiate between reported and ultimate losses. Stating that "losses are decreasing over time" did not receive full credit because it is not clear whether the reported or ultimate losses are decreasing over time.

Part c

In general, candidates performed well on this question.

- By far, the most common mistake that candidates made was to calculate the revised $E(y)$ based on their answers to part (a). A common incorrect answer was $E(y) = 0.8 \times 61.45 = 49.16$. Using the answer from part (a) is not correct as this is an estimate of the *2014 ultimate loss* and not the *total expected ultimate loss*, $E(y)$.
- Candidates generally calculated the value of d correctly. A common mistake was dividing 25 by the answer from part (a) to get $d = 25 / 61.45 = 0.407$.

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

- Other less common mistakes included:
 - Incorrect formulas for VHM and EVPV, including switching the formulas.
 - Not consistently using the values calculated for $E(y)$ and/or d when calculating a revised estimate of ultimate loss for accident year 2014.
 - Using the formula $Z = bd$ to solve for Z , but using the value of b calculated in part (a). This is not correct, since the value of b changes when the expected ultimate loss changes in part (c).
 - Miscellaneous computational errors.